Computational Physics Laboratory Exam – Root Finding

01.
a).
If a function f(x) is **continuous** between two points a and b,
and f(a) and f(b) have **opposite signs**,
then by the Intermediate Value Theorem, there is at least **one root** between a and b.

- If f(a) *f(b) > 0 there is no root between a,b points so then we can't use this method.

- The bisection method finds the **middle point** $c = (a + b)/2$.
- It checks whether the root is between:
  - a and c, or
  - c and b,
    by checking the sign of f(c).
- It **repeats** the process on the half that contains the sign change.

b).

f(a) = f(4) = 64-64+20-2 =18

f(b) = f(6) = 216-144+30-2=100

so , f(a)*f(b) >0 , [4,6] invalid interval choice.

c).

I get [0,3] for interval. 1st time I put larger interval difference like [0,100] then equations show root is near 2. Then in put [0,3] interval for better approximation.

02.

Tolerance	√9 (Bisec(on)	√9 (NewtonRaphson)	√9 (Expected Value)	Absolute Error (Bisec(on)	Absolute Error (NewtonRaphson)
10^{-1}	3.046875	3.00009155413138	3	0.046875	9.15541313801782e-05
10^{-2}	2.998046875	3.000000001396984	3	0.001953125	1.3969838619232178e-09

10^{-3}	2.9998779296875	3.000000001396984	3	0.0001220703125	1.3969838619232178e-09

According to this table shows Newton Raphson method is more accurate than bisection. And also if we minimize the tolerance value then we can improve the accuracy. It valid for both methods. Overall newton Raphson method and it's minimum tolerance value will give the most accurate answer. Newton Raphson method 0.001 tolerance value has minimum Absolute error in this table.